YONGHONG ZHANG

Causal Inference · Reliable AI Agents · Climate Finance

PhD Candidate in Economics & Business, Universidad Autónoma de Madrid · Madrid, Spain

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PROFILE

PhD candidate working at the intersection of causal inference, climate finance, and reliable AI for empirical research. I study whether AI agents genuinely perform execution-critical research tasks and whether their outputs are evaluated against valid criteria. My current work spans agent-benchmark measurement validity, execution-grounded evaluation of causal workflows, evidence-grounded auditing of DID studies, and multi-agent measurement of firm-level ESG perceptions from multilingual news. My empirical research examines the environmental co-benefits of China's emissions-trading pilots using DID and spatial econometrics.

RESEARCH INTERESTS

Causal inference and policy evaluation; reliable AI and agent evaluation; climate finance and carbon markets; firm-level ESG measurement; AI-assisted financial decision-making; evidence-grounded empirical research.

EDUCATION

PhD in Economics & Business, Universidad Autónoma de Madrid — Feb 2024 – Feb 2028 (expected)

Madrid, Spain. Focus: carbon markets, climate finance, environmental policy evaluation, causal inference, AI-assisted empirical research. Supervisors: José Ricardo Vaz Antunes Pereira Correia; Isabel María Parra Oller.

MSc in Corporate & Financial Risk Management, University of Sussex, United Kingdom — Sep 2010 – Feb 2012

Graduate training in risk management, finance, econometrics, and economics.

BSc in Accounting & Business/Management, Brunel University of London, United Kingdom — Sep 2006 – Jul 2010

2+2 articulation: Renmin University of China, School of International Education, Beijing (2006–2008), then Brunel (2008–2010, degree-conferring).

RESEARCH VISITS

Visiting Researcher, Institute of AI in Management, LMU Munich — Sep – Dec 2026

Hosted by Stefan Feuerriegel. Causal design reliability for climate and energy policy evaluation. Funded by an Erasmus+ traineeship grant (KA131-HED) and a Becas Santander predoctoral research grant (UAM).

PUBLICATIONS & WORKING PAPERS

- **CausalVerify: An Execution-Grounded Benchmark for LLM Causal Inference Workflows. Yonghong Zhang**, Ricardo Correia, Isabel M. Parra, and Yong Xie. Benchmarks frontier LLMs on causal-inference workflows, pairing method/design interpretation with execution-grounded coefficient recovery. *Under review. Preprint: arXiv:2609.07944.*
- **Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations. Yonghong Zhang**, Yong Xie, Isabel M. Parra, and Ricardo Correia. Introduces ARGUS, a bounded LLM auditing system for DID identification assumptions using an assumption–implication–evidence rubric. *Proceedings of the 3rd Workshop on NLP Meets Climate Change (ClimateNLP 2026) @ EMNLP, Budapest, pp. 199–219. ACL. aclanthology.org/2026.climatenlp-1.15.*
- **The Double Measurement Confound in Agent Benchmarks: De-Scaffolding, Ground-Truth Scoring, and Reliability Beyond the Mean. Yonghong Zhang** et al. Studies how agent benchmarks can mistakenly measure scaffolds and scorers rather than model capability. *Under review. Preprint: arXiv:2609.09218.*
- **Do Carbon Markets Deliver Environmental Co-Benefits? Evidence from China's ETS Pilots. Yonghong Zhang**, Ricardo Correia, and Isabel M. Parra Oller. Difference-in-differences with province and year fixed effects on a 2004–2022 panel, with spatial-spillover analysis. *Working paper.*

SELECTED RESEARCH & AI PROJECTS

- **GLEAM: Multi-Agent Measurement of Firm-Level ESG Perceptions from Multilingual News.** [Project website] Develops a structured LLM-agent pipeline to construct firm-level environmental, social, and governance perception measures from multilingual news. Ongoing work examines their validity and association with firm-level financial outcomes.
- **ComtradeBench / OpenEnv benchmark.** Deterministic, execution-grounded environment for evaluating

autonomous agents under realistic API frictions; basis for the Web Agent Track and OpenEnv Custom Track results below. Demo: huggingface.co/spaces/yonghongzhang/comtrade-env

HONORS, AWARDS & COMPETITIVE ACHIEVEMENTS

- **1st Place, Best Defender, AgentBeats × Lambda Security Arena 2026,** UC Berkeley RDI × Lambda; Team MateFin; 91.4% defender win rate across 21 held-out adversarial runs.
- **1st Place (tie), Web Agent Track, AgentX–AgentBeats (Phase 1),** UC Berkeley RDI; Google DeepMind-sponsored track; Team MateFin.
- **2nd Place, OpenEnv Custom Track, AgentX–AgentBeats,** UC Berkeley RDI; Team MateFin.
- **Accésit, Best Poster Competition, UAM Doctoral School, May 2026;** one of five awards among 29 posters representing 16 doctoral programmes, decided by attendee vote.
- **Third Prize, 2023 Annual Key Research Project, Rizhao Finance Society, China, awarded April 2024.** Project: Climate Change Risk and Financing Constraints: A Case Study of China.
- **Rising Star of the Year, Shandong Vocational and Technical University of International Studies, 2023.**

EXPERIENCE

PhD Candidate, Universidad Autónoma de Madrid Feb 2024 – present

Doctoral research on carbon markets, climate-policy evaluation, and environmental co-benefits using panel econometrics and causal-inference designs; reproducible empirical workflows in Stata, R, Python-based evaluation workflows, LaTeX/Overleaf, and GitHub.

Lecturer in Financial Management, Shandong Vocational and Technical University of International Studies Sep 2022 – Apr 2024

Taught Corporate Finance and Financial Accounting (bilingual English/Chinese); mentored competition teams; served as an academic reference for students applying to postgraduate programmes in Europe.

Founder & Owner, Bubble-Tea & Coffee Retail Chain, China May 2014 – 2022

Built and ran an independent multi-store chain over ∼8 years, including operations, finance, hiring, suppliers, and local marketing.

English Teacher, Beijing Road Middle School, China 2012 – 2014

Taught English to secondary-school students.

SELECTED PRESENTATIONS

- **Poster, Machine Learning Summer School (MLSS 2026), Tübingen** — presented CausalVerify.
- **ICFAB 2026, Surrey Business School, University of Surrey**, September 2026 — presented the China ETS environmental co-benefits study.
- **Research Seminar, Departamento de Financiación e Investigación Comercial, Universidad Autónoma de Madrid**, June 2026 — presented the China ETS environmental co-benefits study.

METHODS & TECHNICAL SKILLS

Econometrics: difference-in-differences, panel fixed effects, event studies, placebo tests, spatial spillovers, robustness checks. **Machine learning / AI:** LLM agents, retrieval-augmented generation (RAG), evaluation benchmarks, benchmark validity, evidence grounding, agentic workflows, responsible AI. **Research software & tools:** Python-based research and evaluation workflows, Stata, R, Git/GitHub, LaTeX/Overleaf, Hugging Face Spaces. **Data domains:** climate finance, carbon markets, environmental emissions, Chinese provincial panel data, multilingual news, trade/policy datasets. **Communication:** academic writing, research presentations, poster design, English research communication, Chinese-English translation.

TRAINING & SUMMER SCHOOLS

- **Machine Learning Summer School (MLSS 2026, 50th edition),** part of the ELLIS Winter & Summer School Program. Max Planck Institute for Intelligent Systems & University of Tübingen, Germany, 31 August – 11 September 2026. Selected participant; presented a poster.
- **Agentic AI MOOC, UC Berkeley RDI** (Center for Responsible, Decentralized Intelligence): Legendary Tier (top completion tier), Fall 2025.

LANGUAGES

Chinese (native); English (professional working language); Spanish (B1, in progress).